

Triplanes, Gaussians, or Structured Latents? A Comparative Evaluation of Feed-Forward Paradigms for Single-Image 3D Reconstruction

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Abstract—The landscape of single-image 3D reconstruction is currently characterized by rapid yet divergent advances in geometric representations and neural architectures. To navigate this fragmentation, this paper presents a unified, quantitative evaluation of three prominent feed-forward paradigms for zero-shot single-image 3D asset generation: triplane-based Neural Radiance Fields (TripoSR), multi-view 3D Gaussian Splatting (LGM), and Structured Latent Flow Transformers (TRELLIS). This study systematically dissects the fundamental speed-quality trade-offs inherent in these methodologies. TripoSR demonstrates unparalleled inference latency ($<0.5s$) utilizing compact triplane representations. LGM employs multi-view Gaussians, achieving a robust balance of rendering quality and generation speed ($\sim 5s$). Conversely, TRELLIS leverages a structured sparse latent format with rectified flow transformers, yielding high-fidelity, topologically sound meshes at the cost of increased computational latency ($\sim 10s$). These systems are comprehensively assessed across critical metrics, including generation latency, VRAM utilization, Chamfer Distance, F-Score, PSNR, and LPIPS. The empirical findings reveal a distinct Pareto frontier dictating the trade-offs between computational overhead and geometric fidelity, thereby providing a practical framework for deploying 3D generative models in production environments.

Index Terms—3D Reconstruction, Neural Rendering, Triplanes, Gaussian Splatting, Latent Models, Generative AI, Flow Matching

I. INTRODUCTION

The task of reconstructing a fully realized 3D object from a single unposed 2D image is fundamentally ill-posed. A single observation lacks the multi-view geometric constraints necessary for deterministic reconstruction, thereby requiring robust structural priors to hallucinate occluded geometry and complex surface textures. Over recent years, the dominant paradigm for addressing this challenge has shifted significantly. Initial breakthroughs relied on Score Distillation Sampling (SDS), which distilled 2D diffusion priors into Neural Radiance Fields (NeRFs) via per-object optimization [1], [2]. While these optimization-based methods achieved remarkable visual fidelity, they suffered from severe computational bottlenecks and multi-face artifacts, frequently requiring hours of GPU computation per asset.

Consequently, contemporary research has pivoted toward feed-forward architectures. By leveraging massive, curated

3D datasets such as Objaverse [3] to train large-scale neural reconstructors, feed-forward models can infer complete 3D representations in a matter of seconds. However, this acceleration has precipitated an explosion of diverse architectural paradigms. Researchers and practitioners currently face a fragmented array of representational choices, spanning continuous triplane features [4], discrete 3D Gaussians [5], and sparse structured latents [6].

Despite rapid advancements, a critical research gap persists: the lack of a standardized, cross-paradigm evaluation detailing the inherent trade-offs between these fundamentally different representations. This paper addresses this gap by providing a systematic comparative evaluation of three state-of-the-art feed-forward models: TripoSR (representing the triplane paradigm) [7], LGM (representing multi-view Gaussian splatting) [8], and TRELLIS (representing structured latents) [6].

The specific contributions of this work are threefold:

- 1) A comprehensive architectural analysis contrasting the theoretical mechanics of triplanes, multi-view Gaussians, and structured flow latents.
- 2) A rigorous empirical benchmarking across established quantitative metrics (latency, VRAM, Chamfer Distance, F-Score, PSNR, LPIPS), establishing a clear Pareto frontier of computational cost versus reconstruction quality.
- 3) A detailed failure mode analysis identifying the topological and textual limitations inherent to each representation, culminating in a practical deployment taxonomy for downstream applications.

II. RELATED WORK

A. Optimization-Based Reconstruction

The introduction of NeRF [9] established a differentiable volumetric rendering framework that revolutionized 3D synthesis. Subsequent methods, notably DreamFusion [1] and Magic3D [2], leveraged SDS to distill knowledge from pre-trained text-to-image diffusion models into explicit 3D representations. DreamGaussian [10] refined this approach by replacing MLPs with 3D Gaussian Splatting [11], reducing optimization time from hours to minutes. However, the re-

liance on instance-specific optimization remains computationally prohibitive for real-time applications.

B. Multi-View Assisted Generation

To mitigate the inherent ambiguity of single-image inputs, intermediate multi-view generation was introduced. Zero-1-to-3 [12] fine-tuned 2D diffusion models to generate novel views explicitly conditioned on camera extrinsics. Subsequent architectures, such as SyncDreamer [13] and MVDream, utilized 3D-aware attention mechanisms to enforce multi-view consistency. These intermediate representations serve as highly constrained inputs for downstream feed-forward reconstructors.

C. Feed-Forward 3D Models

Large Reconstruction Models (LRMs) [4] demonstrated the viability of pure feed-forward 3D inference using transformer-based architectures that predict triplane features. Splatter Image [5] extended the feed-forward paradigm to directly predict 3D Gaussian primitives. TripoSR [7] advanced the LRM architecture, achieving sub-second inference through optimized channel capacities and sparse rendering logic. Instant3D [14] similarly accelerated generation by combining sparse-view generation with LRM decoders.

D. 3D Foundation Models and Flow Matching

The field is currently transitioning toward generalized 3D foundation models. TRELIS [6] introduced a unified Sparse Latent (SLAT) space for 3D structures, enabling the application of advanced generative techniques such as Conditional Flow Matching [15]. This approach parallels the success of latent diffusion in 2D imagery, currently representing the state-of-the-art in zero-shot geometric and topological fidelity.

III. METHODOLOGY

A. Triplane Neural Radiance Fields (TripoSR)

TripoSR [7] operationalizes the triplane-NeRF paradigm, engineered specifically for real-time feed-forward inference. The pipeline initializes by processing a 512×512 input image through a pre-trained, frozen DINOv1 [16] vision transformer encoder, yielding 1024 latent tokens of dimension 768.

The primary reconstructor is a 16-layer Image-to-Triplane transformer decoder (hidden dimension 1024, 16 attention heads). This transformer utilizes cross-attention to project 2D image features into a discrete set of $32 \times 32 \times 3$ learnable triplane tokens.

These tokens are upsampled and reshaped into a continuous triplane representation $T \in \mathbb{R}^{3 \times 64 \times 64 \times 40}$. TripoSR employs 40 channels per plane—a strategic reduction from the standard 80 channels utilized in base LRMs—to optimize VRAM consumption without severely degrading fidelity. To evaluate a 3D coordinate $x = (x, y, z) \in \mathbb{R}^3$, the point is orthogonally projected onto the XY , XZ , and YZ planes. Bilinear interpolation extracts corresponding features, which are concatenated into a 120-dimensional feature vector $f \in \mathbb{R}^{120}$.

A highly optimized NeRF MLP processes f to output volume density σ and RGB color c :

$$\sigma = \exp(\text{MLP}_\sigma(f) - 1.0) \quad (1)$$

$$c = \text{sigmoid}(\text{MLP}_c(f)) \quad (2)$$

Training is supervised via volume rendering (128 samples per ray). The objective function incorporates a critical mask loss to suppress semi-transparent topological artifacts (floaters):

$$\mathcal{L}_{recon} = \frac{1}{|V|} \sum (\mathcal{L}_{MSE} + 2.0\mathcal{L}_{LPIPS} + 0.05\mathcal{L}_{BCE_{mask}}) \quad (3)$$

Final mesh extraction requires a post-hoc Marching Cubes pass at a 256^3 resolution.

B. Multi-View Gaussian Splatting (LGM)

LGM [8] directly predicts 3D Gaussian primitives using an asymmetric U-Net architecture, circumventing the computational overhead of volumetric ray-marching. To resolve single-image occlusion ambiguity, LGM first utilizes a multi-view diffusion model (e.g., MVDream) to generate four consistent orthogonal views at 256×256 resolution.

To encode spatial relationships and unprojected geometries, LGM projects camera rays using Plücker coordinates. A ray with origin $o \in \mathbb{R}^3$ and direction $d \in \mathbb{R}^3$ is embedded as $(o \times d, d) \in \mathbb{R}^6$. The U-Net input is the concatenation of the RGB color c and the Plücker embedding: $x = [c; o \times d; d] \in \mathbb{R}^9$.

The asymmetric U-Net enforces multi-view consistency through cross-view self-attention layers located at the network bottleneck. The architecture outputs a dense tensor of shape $128 \times 128 \times 4 \times 14$, mapping to 65,536 discrete Gaussians. Each Gaussian is defined by a 14-dimensional parameter vector governing position $\mu \in \mathbb{R}^3$, scale s , rotation quaternion $q \in \mathbb{R}^4$, opacity α , and Spherical Harmonic coefficients $c \in \mathbb{R}^3$.

The model is trained end-to-end via differentiable Gaussian splatting rasterization, optimizing against Ground Truth renderings using combined MSE, LPIPS, and alpha-mask losses.

C. Structured Latent Flow Transformers (TRELIS)

TRELIS [6] introduces a paradigm centered on structured latent spaces, utilizing a Sparse Latent (SLAT) representation. Unlike dense triplanes or unstructured point clouds, SLAT encodes 3D objects as sparse tokens anchored to active surface voxels: $z = \{(z_i, p_i)\}_{i=1}^L$, where $z_i \in \mathbb{R}^8$ is the latent feature and $p_i \in \{0, \dots, 63\}^3$ is the spatial coordinate. Surface sparsity constrains the active token count to $L \approx 20,000$.

Generation employs a two-stage Rectified Flow matching pipeline. Initially, a Sparse VAE encodes binary occupancy $O \in \{0, 1\}^{64^3}$ into structural tokens. Two specialized transformers then govern generation:

- 1) $\mathcal{G}_S$ (**Structure**): A dense transformer utilizing adaptive LayerNorm and DINOv2 cross-attention to predict the coarse bounding structure.

Fig. 1. Visual taxonomy of feed-forward 3D reconstruction outputs. Paradigms range from continuous implicit fields extracted via Marching Cubes (TripoSR), to dense explicit 3D Gaussian primitives (LGM), to highly detailed, differentiable PBR meshes extracted via FlexiCubes (TRELLIS).

- 2) $\mathcal{G}_L$ (**Latent**): A flow transformer that refines local latents z_i , utilizing sparse convolutional packing to maintain tractable sequence lengths.

TRELLIS optimizes a Conditional Flow Matching (CFM) objective, defining linear paths between empirical data x_0 and Gaussian noise ϵ :

$$\mathcal{L}_{CFM} = \mathbb{E}_{t, x_0, \epsilon} [\|v_\theta(x(t), t) - (\epsilon - x_0)\|^2] \quad (4)$$

Crucially, TRELLIS bypasses Marching Cubes by decoding latents via FlexiCubes [17]. The decoder outputs continuous mesh weights and Signed Distance values per voxel, enabling the direct, differentiable extraction of highly detailed 256^3 polygonal meshes.

IV. EXPERIMENTS

A. Experimental Setup

Quantitative evaluations rely on standardized benchmark results aggregated from the original studies [6]–[8], executed on NVIDIA A100 GPUs. Performance is measured using Chamfer Distance (CD) for global geometry, F-Score ($\tau \in \{0.1, 0.2\}$) for surface precision, Peak Signal-to-Noise Ratio (PSNR), and Learned Perceptual Image Patch Similarity (LPIPS) for novel-view synthesis.

B. Quantitative Results

Table I demonstrates the geometric efficacy of the Triplane paradigm (TripoSR) on the Google Scanned Objects (GSO) dataset. TripoSR achieves a Chamfer Distance of 0.111, outperforming baseline LRMs while operating in a fraction of the time.

TABLE I
TRIPLANE (TRIPoSR) EVALUATION ON GSO DATASET [7]

Method	CD ($\downarrow$)	F-Score@0.1 ($\uparrow$)	F-Score@0.2 ($\uparrow$)
OpenLRM	0.180	0.430	0.698
TriplaneGaussian	0.122	0.637	0.846
TripoSR	0.111	0.651	0.871

Table II highlights the performance of the Structured Latent paradigm (TRELLIS) on the Toys4k dataset. The SLAT

representation combined with flow matching achieves near-perfect structural metrics (F-Score 0.999) and superior novel-view fidelity (PSNR 32.74).

TABLE II
STRUCTURED LATENTS (TRELLIS) ON TOYS4K DATASET [6]

Method	PSNR ($\uparrow$)	LPIPS ($\downarrow$)	CD ($\downarrow$)	F-Score ($\uparrow$)
InstantMesh	25.56*	–	0.010	0.964
TRELLIS (GS)	32.74	0.025	0.008	0.999

*Note: Extracted from benchmark subsets. PSNR denotes rendering fidelity.

For the Multi-View Gaussian paradigm (LGM), user studies comparing generations against baselines such as DreamGaussian demonstrate high perceptual satisfaction. Multi-view image consistency was rated at 4.18/5, and overall geometric quality achieved 3.95/5 [8].

C. Computational Efficiency and Hardware Constraints

The defining differentiation among these paradigms is computational efficiency:

- **Triplanes (TripoSR)**: Requires < 0.5 seconds for total inference, consuming 4–6 GB of VRAM. It is the only paradigm viable for real-time, consumer-edge applications.
- **Gaussians (LGM)**: The multi-view diffusion phase requires ~ 4 seconds; U-Net inference requires ~ 1 second. Total VRAM consumption is ~ 10 GB, representing a balanced middle ground.
- **Structured Latents (TRELLIS)**: Deterministic ODE sampling (25–50 steps) requires ~ 10 seconds. Due to sparse voxel tracking and dual transformers, it demands 24 GB of VRAM, restricting it to high-end enterprise hardware.

V. DISCUSSION AND FAILURE MODES

The empirical findings delineate a strict Pareto frontier mapping inference latency against topological fidelity. Triplanes offer real-time speed but are prone to high-frequency detail loss and severe blurring on unobserved object backsides due to continuous field interpolation.

Explicit Gaussians excel in multi-view rendering and texture representation. However, they rely entirely on the accuracy of

the intermediate 2D diffusion step; multi-view inconsistencies manifest directly as floating artifacts. Furthermore, extracting clean polygonal meshes from sparse Gaussian clouds often requires a secondary, computationally expensive volumetric conversion phase, frequently yielding topological holes.

Structured latents provide state-of-the-art geometric fidelity. The integration of FlexiCubes allows for the direct extraction of water-tight, topologically sound meshes without secondary conversions. The primary limitation is computational: high VRAM requirements and longer inference times driven by iterative ODE sampling preclude real-time application. Additionally, the model occasionally bakes directional lighting directly into the surface texture.

TABLE III
COMPARATIVE FAILURE MODE TAXONOMY

Paradigm	Primary Geometric Failure	Mesh Quality	Extraction
Triplanes	Over-smoothing, blurring of unobserved geometries.	Acceptable (Standard Marching Cubes)	
Gaussians	Floating artifacts, thin structure collapse.	Poor (Requires volumetric proxy conversion)	
SLAT (Flow)	Baked-in lighting, high VRAM constraint (OOM).	Excellent (Direct Differentiable FlexiCubes)	(Direct)

VI. CONCLUSION

This study provides a systematic evaluation of contemporary feed-forward 3D reconstruction paradigms. TripoSR proves that sub-second, real-time reconstruction is achievable via optimized triplanes. LGM demonstrates the efficacy of fusing multi-view generative priors with direct Gaussian prediction for high-quality rendering. TRELLIS establishes a new benchmark for topological and geometric quality by applying flow matching to sparse continuous latents. Future research should prioritize hybrid architectures that integrate the rapid inference characteristics of triplanes with the high-fidelity structural priors and differentiable meshing capabilities inherent to structured latent flow models.

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